Power Rule Discovery

Let's see if we can discover a simple rule for finding the ***slopes of power functions****.*

Remember: DERIVATIVES are slopes of curves at one point.

Quadratic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function, $f(x) = x^2$; in the third column are slope values for that function.

x	***f(x) = x²***	***slope or derivative***
-5	25	-10
-3.5	12.25	-7
-1.25	1.5625	-2.5
0	0	0
2	4	4
4.75	22.5625	9.5
7	49	14

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?

Cubic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^3$; in the third column are slope values for that function.

x	***f(x) = x³***	***slope or derivative***
-5	-125	75
-3	-27	27
-1	-1	3
0	0	0
2	8	12
4	64	48
6	216	108

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?

HINT: This pattern is not so easy to detect. Try dividing each of the slopes in the third column by 3.

- Now, do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

Quartic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^4$; in the third column are slope values for that function.

x	*f(x) =x⁴*	*slope or derivative*
-5	625	-500
-3	81	-108
-1	1	-4
0	0	0
2	16	32
4	256	256
6	1296	864

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?

HINT: This pattern is not so easy to detect either. Try dividing each of the slopes in the third column by 4.

- Do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

Quintic

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = x^5$; in the third column are slope values for that function.

x	*f(x) =x⁵*	*slope or derivative*
-5	-3125	3125
-3	-243	405
-1	-1	5
0	0	0
2	32	80
4	1024	1280
6	7776	6480

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

What do you think the rule is to find the slopes (or the derivative) of the function, $f(x) = x^n$?

Verbalizing mathematical content and rules is important. In words, try to state the rule for finding the slopes or the derivative of a power function.

Click on the link in ILearn for the **Derivative App**. To use the app, choose the **quadratic function** and read the directions above the graph. Position the graph so that you can see the whole thing, and go to work on the green dots.

When you have correctly placed all the green dots, you will see a graph and formula for the slopes of the function. Does it match your expectation?

Now, do the **cubic**, **quartic**, and **quintic** functions in the app as well, but don't do any of the others yet. Do your results match your expectations?

Do you think this rule works for other kinds of exponents besides just positive whole numbers?

Let's look at more complicated power functions.

Negative 1 Power

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function, $f(x) = \frac{1}{x}$ *or* $f(x) = x^{-1}$; in the third column are slope values for that function.

x	***f(x) = x*** $^{-1}$	***slope or derivative***
-1/2	-2	-4
-1/3	-3	-9
-1	-1	-1
0	#DIV/0!	#DIV/0!
2	1/2	-1/4
4	1/4	-1/16
6	1/6	-1/36

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

Negative 2 Power

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = \frac{1}{x^2}$ *or* $f(x) = x^{-2}$; in the third column are slope values for that function.

x	***f(x) = x*** $^{-2}$	***slope or derivative***
-1/2	4	16
-1/3	9	54
-1	1	-2
0	#DIV/0!	#DIV/0!
2	1/4	-1/4
4	1/16	-1/32
6	1/36	-1/108

Compare the slopes for the function with *x*-values.

- Do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

Does it look like the rule works for negative powers? Do you suppose it works for fractional powers?

Fractional Power – Square Root

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function, $f(x) = \sqrt{x} \quad or \quad f(x) = x^{1/2}$; in the third column are slope values for that function.

x	***f(x)= x^1/2***	***slope or derivative***
0	0	#DIV/0!
1	1	1/2
4	2	1/4
9	3	1/6
16	4	1/8
25	5	1/10
36	6	1/12

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

Fractional Power – Cube Root

Examine the data table below. In the first column are *x*-values; in the second column are *y*-values for the function $f(x) = \sqrt[3]{x} \quad or \quad f(x) = x^{1/3}$; in the third column are slope values for that function.

x	***f(x)= x^1/3***	***slope or derivative***
-27	-3	1/27
-8	-2	1/12
-1	-1	1/3
0	0	#DIV/0!
1	1	1/3
8	2	1/12
27	3	1/27

Compare the slopes for the function with the *x*-values.

- Do you see a pattern?
- What is the pattern?
- Describe how the numbers in the slope column are related to the *x*-values.

Does it look like the rule works for fractional powers?

Click on the link in ILearn for the **Derivative App**. To use the app, choose the **rational 1 function** and read the directions above the graph. Position the graph so that you can see the whole thing, and go to work on the green dots.

When you have correctly placed all the green dots, you will see a graph and formula for the slopes of the function. Does it match your expectation?

Now, do the **rational 2**, **square root**, and **cube root** functions in the app as well, but don't do any of the others yet. Do your results match your expectations?